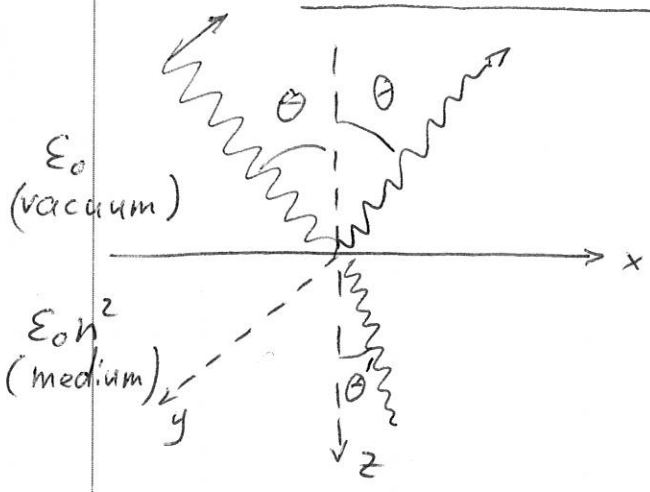
Reflection / refraction of a p-polarized wave



$$\vec{E} = (E_x, 0, E_y)$$

$$\vec{B} = (0, B_y, 0)$$

$$\vec{E}, \vec{B} \sim e^{-i\omega t + i k_x x}$$

$$k_x = \frac{\omega}{c} \qquad k_z = \sqrt{\frac{\omega^2}{c^2} - k_x^2}$$

the same in both media — in vacuum

$$B_y(z<0) = e^{-i\omega t + i k_x x}\left(B_0 e^{i k_z z} + B_r e^{-i k_z z}\right)$$

incident wave — reflected wave

$$B_y(z>0) = B_t e^{-i\omega t + i k_x x} e^{i k_z' z}$$

Continuity at $z=0$:

$$\boxed{B_0 + B_r = B_t}$$

Let us now satisfy the continuity of E_x:

$$\nabla \times \vec{B} = \mu_0 \varepsilon \frac{\partial \vec{E}}{\partial t} = -i\omega\mu_0\varepsilon \vec{E}$$

Taking x-component

$$-i\omega\mu_0\varepsilon E_x = \cancel{\frac{\partial}{\partial y} B_z} - \frac{\partial B_y}{\partial z} = -\frac{\partial B_y}{\partial z}$$

$$-i\omega\mu_0\varepsilon E_z = \frac{\partial}{\partial x} B_y - \cancel{\frac{\partial}{\partial y} B_x} = i k_x B_y$$

$$E_x = -\frac{i}{\omega\mu_0\varepsilon}\frac{\partial B_y}{\partial z} \qquad E_z = -\frac{k_x}{\omega\mu_0\varepsilon}B_y$$

~~This gives for $z<0$ and $z>0$:~~

$$\cancel{E_x = -\frac{k_x}{\omega\mu_0\varepsilon_0}e^{-i\omega t+ik_x x}\left(B_0 e^{ik_z z}+B_r e^{-ik_z z}\right)}$$

Since $\varepsilon E_z = D_z = -\dfrac{k_x}{\omega\mu_0}B_y$

the continuity of D_z is ensured by the continuity of B_y (since k_x and ω are the same in both media)

From the first equation, for the incident wave

$$E_{0x} = \frac{k_z}{\omega\mu_0\varepsilon_0}B_0 e^{-i\omega t+ik_x x+ik_z z}$$

reflected wave

$$E_{rx} = -\frac{k_z}{\omega\mu_0\varepsilon_0}B_r e^{-i\omega t+ik_x x-ik_z z}$$

transmitted wave

$$E_{tx} = \frac{k_z'}{\omega\mu_0\varepsilon}B_t e^{-i\omega t+ik_x x+ik_z' z}$$

Continuity of E_x gives

$$\boxed{k_z\left(\frac{B_0}{\varepsilon_0}-\frac{B_r}{\varepsilon_0}\right) = k_z'\frac{B_t}{\varepsilon}}$$

here $k_z' = \sqrt{\frac{\omega^2}{c^2}\frac{\varepsilon}{\varepsilon_0} - k_x^2} = \sqrt{\frac{\omega^2 n^2}{c^2} - k_x^2} = \frac{\omega}{c}\sqrt{n^2 - \sin^2\theta}$

$$\begin{cases} B_0 + B_r = B_t \\ k_z B_0 - k_z B_r = \frac{k_z'}{n^2} B_t = \frac{k_z'}{n^2}(B_0 + B_r) \end{cases}$$

$$B_r = B_0 \frac{k_z - \frac{k_z'}{n^2}}{k_z + \frac{k_z'}{n^2}} \qquad B_t = B_0 \frac{2k_z}{k_z + \frac{k_z'}{n^2}}$$

$B_r = 0$ when $k_z n^2 = k_z'$

This gives

$$\left(\frac{\omega^2}{c^2} - k_x^2\right) n^4 = \frac{\omega^2 n^2}{c^2} - k_x^2$$

$$\underbrace{(1 - \sin^2\theta)}_{\cos^2\theta} n^4 = n^2 - \sin^2\theta \Rightarrow n^4 = \frac{n^2}{\cos^2\theta} - tg^2\theta$$

Using now $\frac{1}{\cos^2\theta} = 1 + tg^2\theta$ we

find $n^4 = n^2(1 + tg^2\theta) - tg^2\theta \Rightarrow \boxed{tg\theta = n}$

Brewster's angle.

<u>Incidence on the surface of a metal</u>

For simplicity let us consider normal incidence, $k_x = 0$:

$$n^2 = 1 + \frac{i\sigma}{\varepsilon_0 \omega} \approx \frac{i\sigma}{\varepsilon_0 \omega} \quad \text{at small frequencies}$$

Since the conductivity σ usually has both the real and imaginary part (e.g. $\sigma = \frac{ie^2 n}{m(\omega + i/\tau)}$ in the Drude model), refraction index is

complex too.

$$k_z' = \sqrt{\frac{\omega^2}{c^2}n^2 - k_x^2} = \frac{\omega}{c}n \quad (\text{for } k_x = 0)$$

In a special case $\mathrm{Re}\,\sigma \gg \mathrm{Im}\,\sigma$, so that $\sigma \approx \mathrm{Re}\,\sigma$

$$n = \sqrt{\frac{\sigma}{\varepsilon_0\omega}}\sqrt{i} = \frac{1+i}{\sqrt{2}}\sqrt{\frac{\sigma}{\varepsilon_0\omega}} \qquad (*)$$

and $$k_z' = (1+i)\sqrt{\frac{\omega\sigma}{2c^2\varepsilon_0}} = (1+i)\sqrt{\frac{\mu_0\omega\sigma}{2}}$$

This means that the transmitted wave is <u>evanescent</u>:

$$B(z>0) = B_t e^{-i\omega t + ik_z' z} = B_t e^{-i\omega t} e^{i\sqrt{\frac{\mu_0\omega\sigma}{2}}z} e^{-z/\delta}$$

where $\delta = \sqrt{\frac{2}{\mu_0\omega\sigma}}$ is called <u>skin depth</u>

Reflection coefficient

$$R = \left|\frac{B_r}{B_0}\right|^2 = \left|\frac{n^2 k_z - k_z'}{n^2 k_z + k_z'}\right|^2 = \left|\frac{n^2\frac{\omega}{c} - \frac{\omega}{c}n}{n^2\frac{\omega}{c} + \frac{\omega}{c}n}\right|^2 =$$

$$= \left|\frac{n-1}{n+1}\right|^2 = \left|\frac{\mathrm{Re}\,n - 1 + i\,\mathrm{Im}\,n}{\mathrm{Re}\,n + 1 + i\,\mathrm{Im}\,n}\right|^2 =$$

$$= \frac{(\mathrm{Re}\,n - 1)^2 + (\mathrm{Im}\,n)^2}{(\mathrm{Re}\,n + 1)^2 + (\mathrm{Im}\,n)^2} \leq 1$$

where for the real σ, n is given by Eq. (*) above.